Linear Mixed Models

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Restricted maximum likelihood (REML)

Problem: ML estimates of variance components are biased, because we also have to estimate β

REML solution: Write down a new likelihood function for a transformation of **y** that does *not* depend on β

Idea: $y|x \sim N(X\beta, \Sigma)$

$(y - X\beta)|X \sim N(0, \Sigma)$ ← no β!

but requires us to know β?

Idea: if $a \in \mathbb{R}^n$ st $a^T X = 0$

then $a^T X\beta = 0$

$\Rightarrow a^T(y - X\beta) = a^T y$

$\Rightarrow a^T y$ does not depend on β!

Then $a^T y$ is an <u>error contrast</u>

How do we find vectors a st $a^T X = 0$?

REML

$P = X(X^TX)^{-1}X^T$ $\quad PX = X \Rightarrow (I-P)X = 0$

$\Rightarrow$ each entry of $(I-P)y$ is an error contrast!

$\text{rank}(X) = p \Rightarrow \text{rank}(P) = \text{rank}(X) = p$

$\Rightarrow \text{rank}(I-P) = n-p$

Let $C \in \mathbb{R}^{(n-p)\times n}$ be matrix of $n-p$ linearly independent rows of $I-P$

$y|X \sim N(X\beta, \Sigma)$

$\Rightarrow Cy|X \sim N(CX\beta, C\Sigma C^T)$

$(I-P)X = 0 \Rightarrow CX = 0 \Rightarrow CX\beta = 0$

$\Rightarrow Cy|X \sim N(0, C\Sigma C^T)$ ← no β!

Restricted likelihood:

$$L(\Sigma \mid Cy, X) \propto |C\Sigma C^T|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2} y^T C^T (C\Sigma C^T)^{-1} Cy\right\}$$

no β! maximize restricted likelihood to estimate Σ

REML

1. Maximize restricted likelihood to estimate Σ
2. Given $\hat{\Sigma}$,

$$\hat{\beta} = (X^T \hat{\Sigma}^{-1} X)^{-1} X^T \hat{\Sigma}^{-1} y$$

Model assumptions

$$\text{Heterozygosity}_{ij} = \beta_0 + u_i + \beta_1 \log(\text{breeding range}_{ij}) + \beta_2 \log(\text{mass}_{ij}) + \beta_3 \text{latitude}_{ij} + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

Question: What assumptions is this model making?

- constant variance of u_i & ε_{ij} — residual plots!
- Normality of ε_{ij} & of u_i — QQ plots!
- Shape:
  - Linear relationship between explanatory variables & response
  - effects $(\beta_1, \beta_2, \beta_3)$ are the same for each group (family)
- Independence: (groups)
  - families are independent
  - conditional on u_i, observations within groups are independent

residual plots!

Types of residuals

$$y = X\beta + Zu + \varepsilon$$

① $y|X \sim N(X\beta, \Sigma)$

$\hat{\varepsilon} = y - X\hat{\beta}$ (marginal residuals)

② $y|X, u \sim N(X\beta + Zu, Var(\varepsilon))$

$\hat{\varepsilon} = y - X\hat{\beta} - Z\hat{u}$ (conditional residuals)

③ $u \sim N(0, Var(u))$

$\hat{u}$ (random effects)

Problem: $\hat{\varepsilon}$ and $\hat{u}$ are confounded!

Possible solution: if there are enough observations per group, replace $\hat{\varepsilon}$ with least squares conditional residuals (fitting linear regression separately in each group)